

AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Rotor Design of a Novel Synchropter Concept for the Aerial-Recovery of Falling Rocket Stages



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Abstract

From the mission requirements defined for the synchropter in the conceptual design phase, this report investigates and proposes a process that would convert said requirements into some key rotor parameters, to be used as benchmarks for the subsequent rotor design phase. A 7-dimensional design space is defined as possible input space for sizing the synchropter with Blade Element Momentum Theory in both hover and cruise, and two methods are proposed by the author to bypass the computational complexity, one of which is implemented in Isight and MATLAB. Shortcomings of the model used, and also of the process is discussed and will be actioned on in the next design iteration. Optimum rotor design results are presented with a short performance analysis.

This third year Group Design Project (GDP) aims to develop a preliminary design for the mid-air recovery aspect of a wholly reusable launch system for which as much as possible can be recovered and reused for subsequent launches of LEO satellites. This report forms a subset of the marking criteria for the *AERO 96005 - Group Design Project* module, seeking to test engineering judgement and design tradeoffs within group settings.

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

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Nomenclature

α or AoA	Angle of Attack
α_{pitch}	Rotor Pitch Offset Angle
α_{roll}	Rotor Roll Offset Angle
α_{tpp}	Tip Path Plane Angle
$\mathcal{R}$	Blade Aspect Ratio
$\bar{c}_l$	Average 2-Dimensional Lift Coefficient
μ	Advance Ratio
Ω	Rotational speed
ρ	Atmospheric Density
σ	Rotor Solidity
θ_0	Collective Angle in Cruise
A	Disc Area
a	Airfoil 2-Dimensional Lift Curve Slope
c	Chord
C_P	Coefficient of Power, N.B. numerically equal to C_Q if defined in NASA convention
C_Q	Coefficient of Torque
C_T	Coefficient of Thrust
$C_{L_{max}}$	Stall Lift Coefficient
$D.L.$	Disc Loading
FM	Hover Figure of Merit
k	Collective Angle in Hover
k	Factor to account for Overlap in Rotor Area
M	Mach Number
M_{adv}	Advancing Blade Mach Number
M_{DD}	Drag Divergence Mach Number
$MTOW$	Maximum Takeoff Weight
n	Number of Blades in a Rotor
P	Power
R	Blade Radius

T Thrust

T_C Thrust in Cruise

T_H Thrust in Hover

$taper_1$ Root Linear Taper

$taper_2$ Tip Linear Taper

$trans$ Airfoil Spanwise Transition Location

$twist_1$ Root Linear Twist (negative)

$twist_2$ Tip Linear Twist (negative)

V or V_∞ Forward Speed

$\frac{L}{D_{heli}}$ Lift over Drag Ratio of entire Helicopter

Part I

Introduction

This report aims to take the reader through the work done from mission requirements to the final rotor design, and references to other reports by different team members are made to give the reader more of a complete picture of the design process and thinking. We will start from translating mission requirements to key rotor parameters, which will be used together with a set of geometric parameters describing the blade planform in the in-house BEMT code. Optimisation will ensue, and the final rotor design will be presented. As the subteam coordinator, the division of labour to deliver the rotor design component of the Aerial Recovery Project within the Aerodynamics team is also touched upon.

Part II

Design Choices of Key Rotor Parameters

Once the synchropter configuration is given the go-ahead, there are restrictions on the rotor design posed by the mission requirements, which we must translate to a list of target rotor parameters.

1 Number of blades

Number of blades on a rotor was chosen to be 2 in the conceptual design phase, as any other number of blades will increase the difficulty of the meshing of the synchropter.

2 Disc Loading

Disc loading, defined by the thrust, T , divided by the disc area, A , is arguably the most definitive parameter of the initial sizing of a helicopter. It dictates the disc area, and hence blade radius, given a thrust requirement, and characterises the efficiency and induced velocity of the rotor. Qualitatively, the higher the disc loading, the more compact and lightweight the helicopter will be (lower empty weight), and the more operational challenges it may present. The operational challenges stem from the fact that a higher disc loading brings a higher induced velocity in the wake of the helicopter, and this in turn makes it

- Difficult to work under the hovering rotor for recovery.
- Likely to lift up debris on the surface to obstruct takeoff and landing.

With disc loading of conventional helicopters usually lying within the range of 3-12 lbf/ft². After much discussion with Thomas [1] and Omar [2] from the Systems and Propulsion team, it is decided initially that a disc loading in the lower region of the range, 6.0 lbf/ft², would be appropriate for the mission.

There is then the ambiguity of the definition of the disc loading, which unfortunately is only discovered further into the project. For a synchropter the total helicopter disc loading, defined by the total thrust generated over the total disc area, is different from the individual rotor disc loading, which only considers the per rotor values, due to the fact that the two rotors operate over an overlapping area, which must be discounted from the total disc area calculation. To preserve the rotor parameters which have already been sized for, a difficult decision is hence made [3] to raise the total disc loading to 9.4 lbf/ft² to keep the individual rotor disc loading the same at 6.0 lbf/ft².

3 Blade Radius

The blade radius is an important parameter in the weight sizing of the helicopter, as the weight of the blades, and more importantly the lift it can produce, are key inputs to the iterative process for finding the MTOW of the vehicle [4] [5]. Given MTOW from the previous iteration, blade radius, R , is related by

$$k\pi R^2 D.L. = MTOW, \quad (1)$$

where k is the factor to account for the total disc area ($k = 1$ for coaxial rotors, $1 < k < 2$ for synchropter, $k = 2$ for tandem rotors), and in the Imperial unit system, R is in ft, $D.L.$ is in lbf/ft² and $MTOW$ is in lbf.

An earlier calculation which erroneously cited $k = 2$, i.e. due to the confusion of total disc loading and individual rotor disc loading as stated earlier ($D.L. = 6$), led to an underestimated blade radius of 13.2 ft for a converged $MTOW$ of 2907kg. The blade radius is then decided to be retained to minimise changes which would cascade down to the other teams working on the empennage, fuselage, etc. by raising the total disc loading to 9.4 lbf/ft² as discussed earlier for an estimated k of 1.25 based on the rotor separation distance as given by the Systems and Propulsion team [1].

4 Solidity/Aspect Ratio

For a fixed blade radius, the solidity σ , is given by the ratio of blade to disc area, assuming no taper

$$\sigma = \frac{ncR}{\pi R^2} = \frac{nc}{\pi R}, \quad (2)$$

where n is the number of blades and c is the (constant) chord length. It can be shown that the aspect ratio (the aircraft equivalent) is effectively another measure of solidity by considering

$$\mathcal{R} = \frac{(2R)^2}{2cR} = \frac{2n}{\sigma\pi}. \quad (3)$$

However, solidity is not merely a geometric measure. An analogy was made [6, P. 17] to compare C_T/σ in helicopter aerodynamics to C_L , where C_T is the helicopter coefficient of thrust (note the lack of the 1/2 factor, the NASA system is used throughout this report)

$$\frac{C_T}{\sigma} = \frac{T}{\rho\sigma(\pi R^2)(\Omega R)^2}, \quad (4)$$

and where Ω is the rotational speed of the rotor, Hence, we compare C_L to $C_{L_{max}}$ to detect stall, likewise we compare C_T to a defined maximum. To explore the relationship we define the lift per running foot in hover by $\bar{c}_l$, the average 2-D lift coefficient along the blade

$$\frac{\Delta L}{\Delta r} = \frac{1}{2}\rho(\Omega r)^2 \bar{c}_l c, \quad (5)$$

which when integrated along the blade and multiplying by the number of blades gives

$$T = n \frac{1}{2} \rho \Omega^2 \bar{c}_l c \int_0^R r^2 dr = \frac{1}{6} \rho (\Omega R)^2 n c R \bar{c}_l, \quad (6)$$

and finally combining with Equation 4 gives

$$\frac{C_T}{\sigma} = \frac{1}{6} \bar{c}_l \quad (7)$$

which then we realise for a given MTOW requirement (on C_T), there is a minimum solidity requirement to not exceed the $C_{L_{max}}$ of the airfoil used. Recalling equation 2 that for a given blade radius, the lower the solidity, the shorter the chord length, and the lighter the blade.

With conventional rotor blade airfoils, it is reasonable to assume their $C_{L_{max}}$ would lie in the range of 1.0 to 1.4, and hence $\bar{c}_l$ to be within the range of 0.6 to 0.8, which for a target C_T of 0.010, gives a range of $\mathcal{R}$ of 13 to 17. An aspect ratio of 14.0, corresponding to a solidity of 0.091 is chosen after much discussion with the Structures and CAD team [7], mainly a compromise between structural integrity especially at the root with an increasingly slender blade, versus the gain in aerodynamic efficiency at a higher aspect ratio.

5 Tip Speeds and Rotational Speed

The speed at which the tip rotates in hover is constant, however in forward flight that will not be the case. Hence the advancing blade tip speed, at the azimuthal location where the blade rotational speed and forward speed vector points in the same direction, will be used to characterise the speed in forward flight. This speed is limited by the onset of compressibility effects at the tip due to the high local Mach number, but it can be mitigated by the use of an airfoil with a high drag divergence Mach number M_{DD} , together with sweepback at the tip section of the blade. There is indeed again another tradeoff to consider, as roughly speaking the higher the tip speed, the higher the dynamic pressure and hence the higher the lift generated, if we are fixing the spanwise angle of incidence by appropriate use of blade twist. Combining with the fact that we have to size the rotors for a maximum dash speed of 225 km/h at 20000 ft, we have decided that an advancing tip Mach number, M_{adv} of 0.85 would be appropriate, which assuming a constant blade rotational speed regardless of flight regime (hover or forward) would give us a hover tip Mach number, $M_H = 0.65$ (also at 20000 ft).

As for the rotational speed, there is indeed the option to increase the blade rotational speed in hover, to generate some extra lift due to the fact that $M_H = 0.65$ is almost certainly below the M_{DD} of the airfoil used for both the tip and root sections. However, that would require a gearbox system in the rotational shaft of both rotors, and after consulting the Systems and Propulsion team, it is decided that the weight penalty and design complexity of such a system would outweigh the potential benefit in having an additional input variable for the thrust/lift output.[1] Hence, the decision was made to keep the rotational speed constant in all flight regimes, at 8.13 rev/s.

6 Summary of Key Parameters

The parameters in the above sections hence becomes the target specifications in the preliminary blade design process. They will be achieved with the right combination of blade parameters and airfoil choices.

Table 2: *Rotor target specifications*

Quantities	Values	Assumptions
Number of Blades, n	2	(chosen)
Total Disc Loading	9.4 lbf/ft ²	$k = 1.25$
Rotor Disc Loading	6.0 lbf/ft ²	(chosen)
Blade Radius, R	13.2 ft	(convergence after iteration)
Solidity, σ	0.0909	constant chord
Aspect Ratio, $\mathcal{A}$	14.0	constant chord
Advancing Blade Mach number, M_{adv}	0.85	225 km/h at 20000 ft
Hover Mach number, M_H	0.65	at 20000 ft
Rotational Speed, $\frac{\Omega}{2\pi R}$	8.13 rev/s	constant in all flight regimes

Part III

Preliminary Design and Optimisation of Rotor Blade

To achieve the tabulated rotor specifications, it is trivial that just using one airfoil section for the entire rotor blade is not ideal, since we have conflicting design drivers at the blade root and tip. At the blade root, an airfoil section with a higher thickness to chord ratio is desired because usually

- the stall AoA will be higher to delay separation given the high angle of attack.
- the $C_{L_{max}}$ will be higher to compensate the low local flow speed seen to provide more lift.
- the stall behaviour will be more benign and prevent the loss of lift altogether.
- the increased thickness will withstand better the high torsional and bending loads at the root.

And at the blade tip, an airfoil section with a lower thickness to chord ratio is desired because usually

- the M_{DD} will be higher to withstand or avoid altogether compressibility effects.
- the decreased thickness will impart less torsional and bending loads on the blade structure.

A detailed quantitative analysis is conducted by Alessandro [8] and Kevin [9], which involved short-listing candidate airfoils, computing their lift and drag polar with CFD methods, and finally deciding on the appropriate airfoil for each section. It is decided that the **NACA23012** will be used for the root section, and the **NACA63206** will be used for the tip section.

7 Qualitative Understanding of the Design Space

With the two airfoils selected, to fully define the geometry of the two-section planform we introduce the following parameters.

Table 3: *Parameters for Blade Design*

Quantities	Symbol	Range	Definition
Airfoil Spanwise Transition	$trans$	0 to 1	spanwise non-dimensional distance
Root Linear Twist (negative)	$twist_1$	0 to ~ 15	degrees relative to root
Tip Linear Twist (negative)	$twist_2$	$twist_1$ to ~ 20	degrees relative to root
Root Linear Taper	$taper_1$	0 to 1	non-dimensional chord length
Tip Linear Taper	$taper_2$	$taper_1$ to 1	non-dimensional chord length
Collective Angle in Hover	k	~ 10 to ~ 25	degrees required to trim
Collective Angle in Cruise	θ_0	~ 10 to ~ 25	degrees required to trim

N.B. the twist parameters are defined such that at the root, twist is 0 degrees, and at the end of the root ($r/R = trans$) the local angle of twist is $-twist_1$, etc. and the collective angle is a constant added to the spanwise linear twist distribution such that it is a fully defined linear distribution ($y = mx + c$ rather than $y = mx$) so the local blade incidence can be controlled (collective pitch) to match the local flow conditions to maximise the lifting efficiency. Note that the local flow velocity distribution is almost always non-linear, and the bi-linear twist approach is an attempt to achieve an acceptable twist distribution without overcrowding the design space with variables (and increasing the difficulty of manufacture of course). Similarly with the linear taper to redistribute the lift, a bi-linear model is chosen to reduce the complexity while capturing the effects of such a tradeoff in the optimisation process.

It is understood that there are optimal points for hover and cruise respectively achieved by the variation of the above to create unique planforms, and the optimal planform for the blade design would be a compromise of two parameters characterising hover and cruise performance.

Specifically, for $trans$, the optimum should be near 1 because the transition of airfoil should be as far spanwise as possible to reap the superior lifting characteristics of the root airfoil at low Mach numbers, while not too far to also reap the good wave-drag resistance of the tip airfoil at higher Mach numbers.

For twist, a perfect twist distribution for hover would keep the local angle of incidence at $AoA_{maxL/D}$ of the airfoil at all spanwise location to achieve max L/D overall, and that twist distribution would be a hyperbolic distribution, $twist(r/R) = \frac{twist(1)}{r/R}$. [10, P. 128]. However, in forward flight, excessive amounts of twist at the tip may bring the local angle of incidence to near zero or even negative values, compromising the cruise performance. [10, P. 290]

For chord, a perfect distribution for hover would like in a similar form to twist, $chord(r/R) = \frac{chord(1)}{r/R}$. [10, P. 136] However, in forward flight, the sweep that would arise because of taper would be beneficial in reducing the local Mach number and improve high speed performance at the advancing tip, where in the ideal taper distribution for hover, at the tip the chord is nearly constant.

k and θ_0 are completely decoupled parameters, one for each flight regime to optimise (hover, cruise). They are there simply to ensure the defined planform can achieve the required thrust/lift without stalling.

8 Theory and Formulation of Optimisation Problem

There are three main families of methods to analyse the aerodynamic performance of a rotor blade, in increasing order of complexity

1. Momentum/ Actuator Disc Theory, which treats the rotor as a homogenous disc.
2. Blade Element Momentum Theory (BEMT), which treats the rotor blades as discretised elements in radial and azimuthal directions.
3. Free Vortex Wake Theory, which instead of assuming a form of the inflow (often linear or constant in above theories), calculates the exact inflow distribution by solving systems of equation describing the characteristics and interactions of the vortexes shed by the tip during rotation.

It is perhaps unsurprising that the implementation of the free vortex wake theory is too computationally expensive for the purpose of this project, and hence to relate the aforementioned blade geometry (input) parameters to the performance (output) parameters of the blade, BEMT is chosen as the method to go with. Enson [11] has completed the analysis from first principles, while myself relied extensively on the implementation described in [6, P. 69, 209]. I will hence briefly lay out the implementation and verification of the BEMT code that is developed independently from Enson.

8.1 Theory and Implementation of the BEMT

8.1.1 Hover

The direct process described in Chapter 1 of Prouty[6, P. 69] is followed for hover at 20000 ft with MTOW and results are presented and compared with the test case provided in Figure 1.

The results are mostly in line with the established results, except a slight deviation in the local lift curve slope of the airfoil along the blade, due to the raw data not being available. That is also the single biggest cause of the discrepancies in the calculated constants.

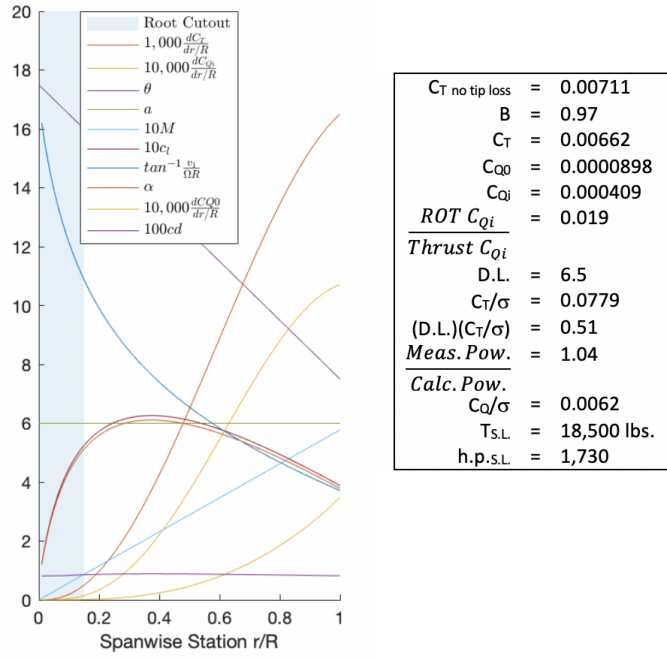
The code for hover flight takes in the key rotor parameters and the geometry parameters to give the thrust and power of the rotors, individual rotor disc loading, and the Figure of Merit, FM, defined by

$$FM = \frac{C_T^{3/2}}{\sqrt{2}C_Q} \quad (8)$$

8.1.2 Cruise

The iterative process described in Chapter 3 of Prouty[6, P. 209] utilising a linear inflow assumption is followed for a forward dash speed of 225 kmh^{-1} at 20000 ft with MTOW as in the mission requirements and converged results are presented and compared with the AoA contour plot provided in Figure 2.

The shape of the contours are comparable in both of the plots, except perhaps the region of reversed flow, where it is modelled slightly bigger than in the provided plot. As the local velocity at the region of reversed flow is usually low, the discrepancies in calculated thrust/lift quantities will be insignificant.



(a) Results with inhouse code

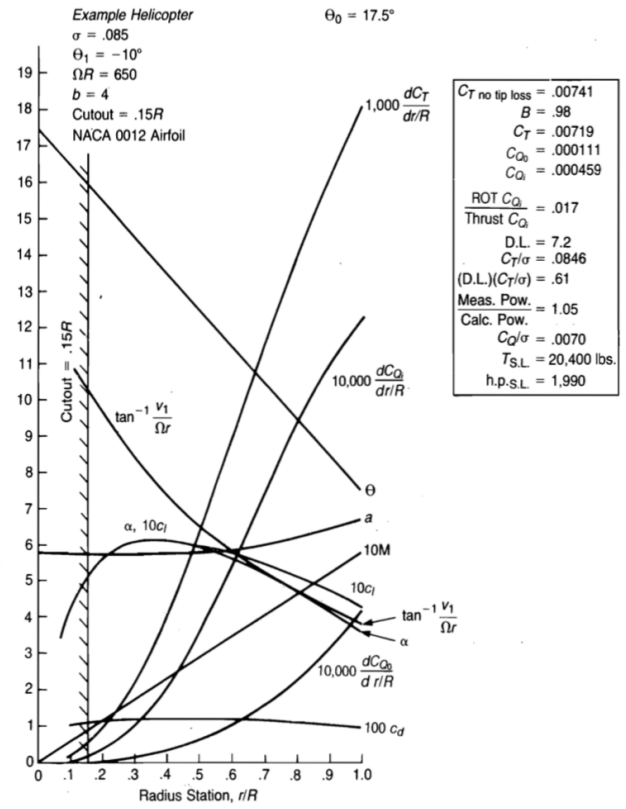
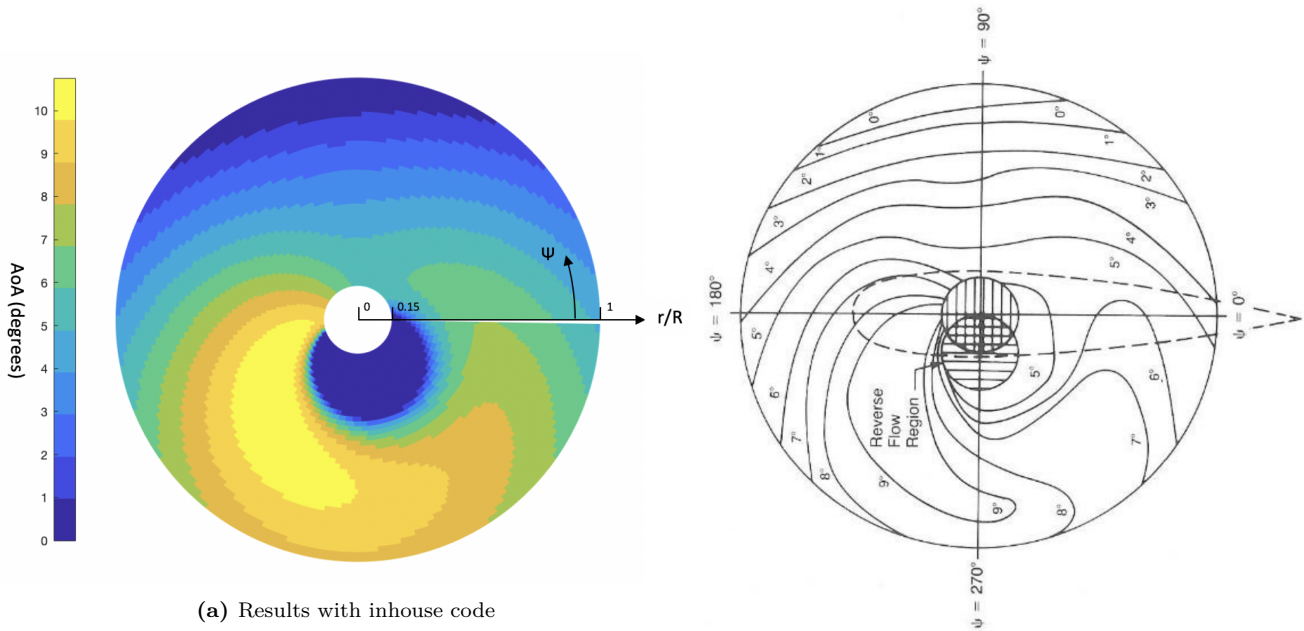


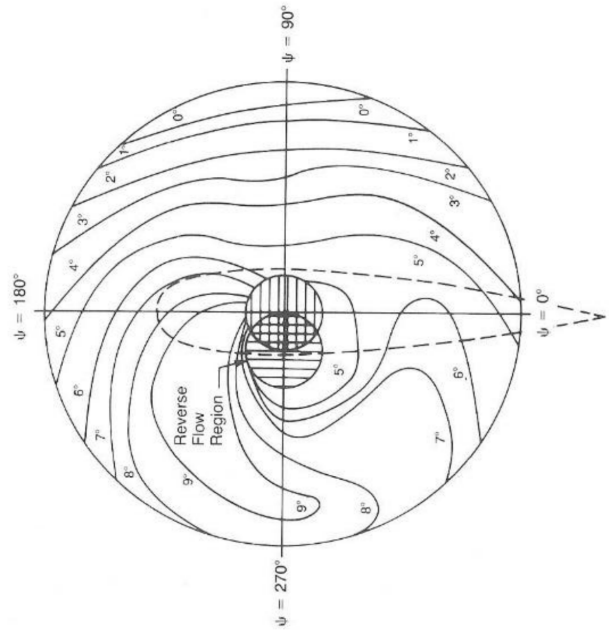
FIGURE 1.45 Results of Sample Calculation

(b) Results from Prouty[6, P. 76]

Figure 1: Comparison of results for validity in hover flight



(a) Results with inhouse code



(b) Results from Prouty[6, P. 162]

Figure 2: Comparison of results for validity in cruise flight

The code for cruise flight takes in the key rotor parameters and the geometry parameters, and also an additional helicopter drag parameter, the equivalent flat plate area, f , which is calculated in the helicopter drag estimation by Alessandro [8], to give the thrust and power of the rotors, and the helicopter lift over drag ratio, L/D_{heli} , by

$$\frac{L}{D_{heli}} = \frac{T \cos(\alpha_{tpp}) \cos(\alpha_{pitch}) \cos(\alpha_{roll})}{P/V_\infty} \approx \frac{WV_\infty}{P} [10, P.229], \quad (9)$$

where α_{tpp} , an iterated output, is the angle of the tip path plane with respect to the undeformed rotor tip plane when the rotor is producing lift, $\alpha_{pitch} = 3 \text{ deg}$ and $\alpha_{roll} = 14 \text{ deg}$ is the tilt of the rotor shaft in pitch and roll direction respectively, determined by the Systems and Propulsion team[1].

8.2 Optimisation Problem

We have one more parameter, the root cutout, to determine before we have all the inputs required to define the optimisation problem. Fortunately, the root cutout is dependent on the ability to house the shaft and other related turbomachinery sub-assemblies in a small hub [1], and at the same time the structural implications at the transition of the blades from the hub [7]. After a joint meeting of the three relevant parties it is decided that $r/R=0.15$ is a good compromise for the root cutout, to not sacrifice too much lift while still reserving enough margin for the volume and structural integrity of the hub.

Having completed the set of inputs for the optimisation process, we conclude that the 7 geometry parameters in Table 2 can vary in their own ranges to form a 7-dimensional design space, where each design case would come with the thrust in hover and cruise, D.L. and FM in hover, and $\frac{L}{D_{heli}}$ in cruise. The thrust in hover and cruise together with the D.L. are used to match cases with the target thrust (using the MTOW) and disc loading (prescribed individual rotor D.L.), and those cases with thrust and D.L. within a certain target range will be plotted with their FM against $\frac{L}{D_{heli}}$, to identify the design which presents itself as a good compromise of both hover and cruise performance.[12]

9 Simplification and Branching

A naive attempt to find the global optimum for the above problem would be to brute force through all possible combinations of the parameters. However, if we use a discretisation of 10 in each of the possible range, and consider a runtime of 0.1 s in MATLAB for each case to converge, that would amount to 11.6 days of runtime which is not feasible. Clearly, there is the need to either decouple the system with the right assumptions, or find the global maximum with a more efficient algorithm.

9.1 Decoupling

With the former approach, considering the fact that the root airfoil is superior than the tip airfoil in practically every scenario except at high Mach numbers, a reasonable engineering decision was made to set $trans = 0.9$ to leave sufficient room for the tip transition and design. Taper in helicopter blades is beneficial to hover performance, but the gain in FM with an ideal taper distribution is only 5-6% [10, P. 154], and can be achieved with appropriate taper at the tip section alone. Hence we set $taper_1 = 1$, and are left with five free parameters. The design of the tip is now not dependent on any free parameters except twist, and can be decoupled from the root design. As the coordinator of the Aerodynamics team, I have delegated the tip design process to size for the tip sweep and chord/planform ($taper_2$) to Kevin [9], and the optimisation of the remaining four parameters to

Enson [11], given the order of magnitude of the complexity has reduced by 3 and we can achieve acceptable resolution of the design space.

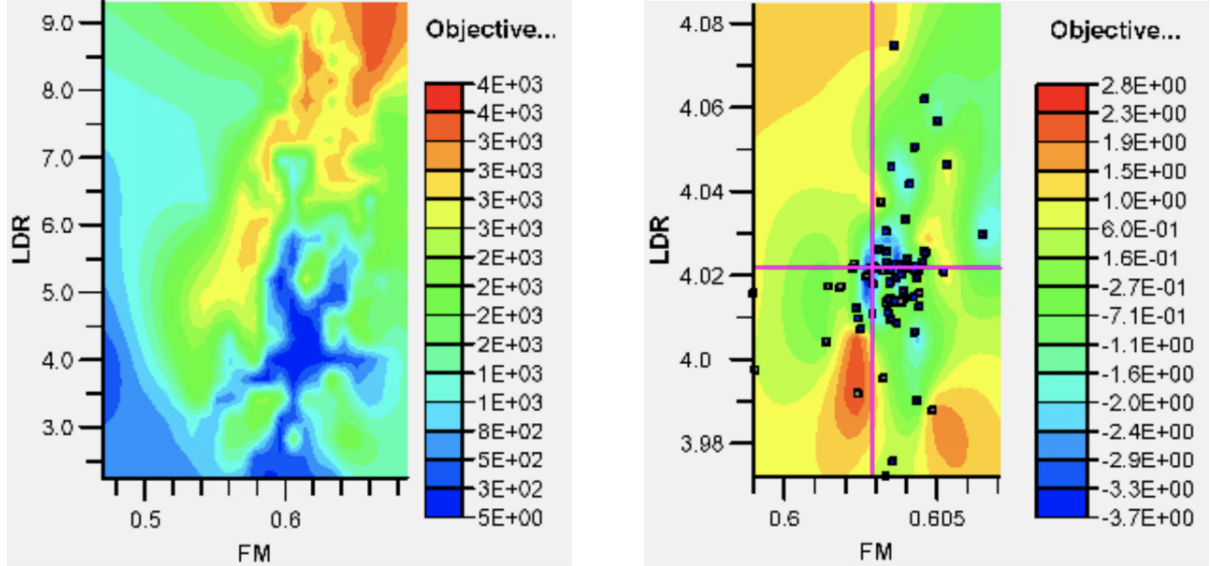
9.2 Analysis with Isight

At the same time, I myself had taken up the task of migrating from brute-forcing with MATLAB to a more efficient optimisation process. Courtesy to Dr. M. Ribera, Department of Aeronautics, who has kindly introduced me to the optimisation workflow in Isight, I was able to conduct the optimisation of the full 7 parameter problem. For the setup please refer to Figure 5 in the Appendix.

9.3 Algorithm Selection

Adaptive simulated annealing was chosen as the appropriate algorithm for the purpose of this optimisation task. The algorithm will look for a global minimum for a cost/objective function, which is suitably defined by objectives. It is analogous to a bouncing ball in a mountainous region, which would bounce over mountaintops given it possesses enough energy. As the energy of the ball reduces, it would bounce in smaller ranges of valleys. The cost/objective function of the alleys are then compared, and we will continue bouncing in the valley with the lowest cost/objective function.[13] In our case, if the optimum is indeed a tradeoff between FM and $\frac{L}{D}_{heli}$ we would be able to visualise the curved trench of the two variables in the contour plot of the cost/objective function.

9.4 Results from Isight



(a) Contour plot of objective function for all data points (2524, points hidden) investigated for $\frac{L}{D}_{heli}$ vs FM

(b) Contour plot of objective function for candidate points (101) for $\frac{L}{D}_{heli}$ vs FM

Figure 3: Contour plots of objective function for $\frac{L}{D}_{heli}$ vs FM from Isight

Figure 3 shows the contour plots obtained by Isight with the above-mentioned inputs and algorithm. It can be seen in Figure 3a that the global minimum of the objective function is not a trench, and instead is a point somewhere at FM= 0.6 and $\frac{L}{D}_{heli} = 4.0$. Just plotting the points with a low objective function (-4 to 4) gives Figure 3b. The highlighted point is the global optima obtained by Isight, with a FM of 0.603, and $\frac{L}{D}_{heli}$ of 4.02.

Table 4: *Isight Optimised Blade Design Parameters*

Quantities	Value
trans	0.90
twist_1	10.92 degrees
twist_2	18.01 degrees
taper_1	0.567
taper_2	0.768
k	24.40 degrees
θ_0	20.62 degrees

By inspection this blade planform would not be ideal in terms of structural integrity, because the hourglass taper along the span would introduce a weak point at $r/R = 0.9$. By considering the discussion on taper in Section 9.1, we have set $taper_1 = 1$, and the fact that a BERP planform is decided to be used at the blade tip [9] means that further discussion on $taper_2$ is irrelevant in this report.

10 Iteration and Final Design Decision

The optimised parameters are then compared with the results obtained in MATLAB by Enson [11] and they are in good agreement. The case is also checked with the stall criterion to ensure no part of the blade except that in the region of reversed flow will stall.

Table 5: *Rotor Design Results (per rotor)*

Quantities	Value
trans	0.90
twist_1	11.0 degrees
twist_2	17.0 degrees
taper_1	1
taper_2	see [9]
k	24.0 degrees
θ_0	22.0 degrees

Internally, these parameters are passed on to Kevin[9] and Alessandro[8], whose work on tail and empennage design is dependent on the results of the rotor design. They have worked closely with the Flight Mechanics and Structures teams in the design process.

These parameters are also passed on to the Structures and CAD team, whom after some structural sizing, confirmed that it is feasible and will be built out of carbon fibre[7]. The maximum thickness of the blade would be 3 mm , which is reasonable and appropriate. The blade sub-assembly will be then modelled in X-Plane 11 by the Simulations team to perform flight testing and verify that it meets the target specifications. An acoustic study is also performed by Alessandro [3] and that might also have an impact on the rotor design.

Part IV

Performance Estimation

11 Maximum Forward Speed without Payload

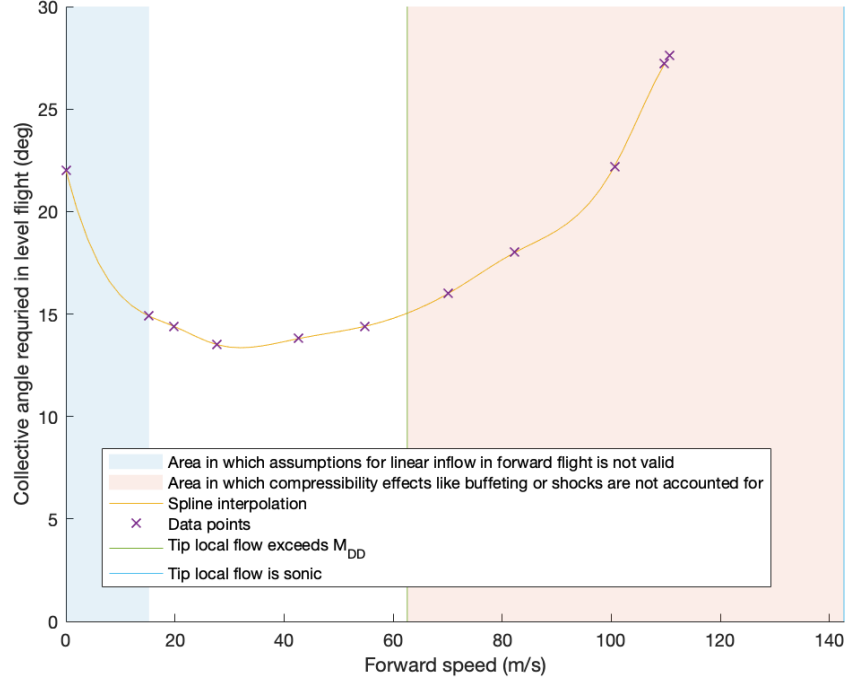


Figure 4: Estimation of maximum forward flight speed at 20000 ft with no payload

To produce Figure 4, different values of forward speed are chosen and the M_{adv} is calculated and fed into the script, to iterate for the collective angle that would trim the helicopter. The script was unable to achieve trim at any forward speed $0 < V < 15.2 \text{ ms}^{-1}$ (blue region), reason for which will be presented in Part V. Caution must be taken to interpret any data in the red region, as in this region the local tip Mach number has exceeded M_{DD} of the tip airfoil, and compressibility effects like buffeting and potentially shocks are not modelled. At $V > 110.6 \text{ ms}^{-1}$ the code will not converge, which is due to stall of the entire blade, when any increase in collective angle cannot provide the lift required to support the weight. Based on the above, we cannot possibly determine a maximum forward speed without a more complete, perhaps 3D analysis of the blade in forward flight. The Simulations team [14] will perform flight tests on the helicopter to put a definite number to this metric.

12 Other Performance Metric

For some particular performance metric predicted by BEMT of the finalised blade design please refer to Table 6 in the Appendix. For other rotor-related performance metric please refer to [11] and [15].

Part V

Discussion and Conclusion

13 Why is the optimisation result from Isight not a curve?

Contrary to our expectation, the results we obtained from Isight is rather unlike a pareto front. It is partly due to the limitation of our code - the inability to prescribe a D.L. or thrust requirement together with the geometry parameters, and instead requiring the filtering of cases to compare configurations that are able to produce the same thrust and D.L. simply means out of the cases considered by Isight, only a small number of them are useful in plotting the pareto front. It might also be partly due to the combination of the optimisation algorithm and setup of the objective function, which unfortunately cannot be verified at this stage due to the limited time.

14 Why does the cruise code not converge below a certain V?

That would be due to the fact that a linear inflow assumption for forward flight is used, which introduced a function of the inverse of the advance ratio, μ , defined by $\mu = \frac{V}{\Omega R}$, in the lateral flapping equation. As a result, the lateral flapping will approach infinity when the helicopter slows down to approach hover. The lower limit to advance ratio where results obtained are still acceptable would be at $\mu > 0.05$ [6, P. 173], which would translate to $V > 10.3 \text{ ms}^{-1}$.

15 How does the helicopter being a synchropter affect the results?

In the BEMT code it is assumed inflows of rotor will not affect each other, that is, essentially they are two separate rotors, each contributing lift to hoist half of the MTOW (at least in hover), acting infinitely far away from each other. Literature on synchropters are few and far between, and there is no easy theory to account for the interaction of the wake. However, there exist a correction factor derived using momentum theory which applies in very specific conditions of the rotor layout [10, P. 106], which will be applied in the next iteration. The total helicopter disc loading is where the synchropter effect is being considered in the current model.

16 How can we improve in the next iteration?

We should confirm the target specifications are met with the current blade design after verification in X-Plane 11, and reconcile discrepancies should any arise. Tip design should be also included in the quantitative analysis, in particular, how the increased mass at the tip will affect the performance of the rotor. If time allows, we should migrate to a free wake vortex theory based model, to achieve higher fidelity of results at low advance ratios, to better account for the effects of a synchropter, and to eliminate any assumptions of the inflow in the calculations. [16] We should also rewrite the code such that thrust and disc loading are included as prescribed parameters to be automatically sized to, instead of relying on filtering of cases.

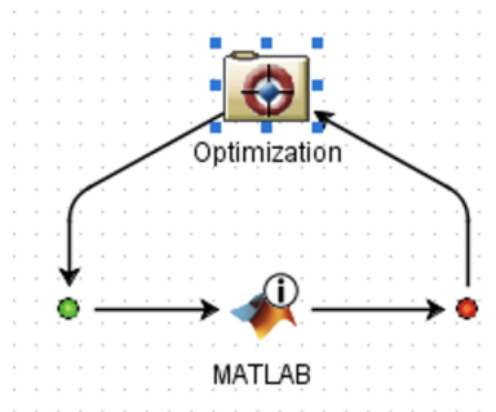
17 Conclusion

Hence, using BEMT to build a model of the rotors in hover and cruise we are able to arrive at an optimum rotor design case by solving the optimisation problem using Isight and MATLAB. The entire design process, the theory and implementation, is proposed, debated, and proven to work. Potential improvements, both to the model and the process is discussed.

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- [15] K.S. Hon. Unmanned Helicopter for Aerial Recovery - Individual Report. 2020. WRR Group Design Project, Flight Mechanics, Performance, and Control.
- [16] M. Ribera. Helicopter Flight Dynamics Simulation With A Time-accurate Free-vortex Wake Model. 2020.

A Isight Setup



(a) Work flow setup in Isight

Component Editor - Optimization

Optimization

General Variables Constraints Objectives

Parameter	Lower Bound	Value	Upper Bound	Allowed Values	Scale Factor
blade_c_ratio_1	0.5	0.9	1.0		
blade_c_ratio_2	0.5	0.9	1.0		
blade_twist_1	7.0	10.0	15.0		
blade_twist_2	7.0	10.0	20.0		
k	-25.0	-15.0	-14.0		
thetanaught	14.0	20.0	25.0		
trans		0.9		0.1;0.2;0.3;0.4;0...	

(b) Parameters setup in Isight, note slight difference in nomenclatures, range decided by experience

Component Editor - Optimization

Optimization

General Variables Constraints Objectives

Parameter	Direction	Target	Scale Factor	Weight Fac...
DL	target	6.36	1.0	1.0
FM	maximize		1.0	1.0
L	target	3296.0	1.0	1.0
LDR	maximize		1.0	1.0
Th	target	3296.0	1.0	1.0

(c) Objectives setup in Isight, note slight differences in nomenclature

Figure 5: Isight Setup

B Prediction of Performance by BEMT

Quantities	Value
T_H	3510 lbf
T_F	3090 lbf
Individual Rotor $D.L.$	6.41 lbf/ft ²
FM	0.610
$\frac{L}{D}_{heli}$	3.40

Table 6: Performance Parameters Predicted by BEMT

C MATLAB Code for Hover

This is the implementation of the hover BEMT.

```

1
2 function [T, hp, maxaoa, DL, FM]=hover(blade_num, blade_r, sigma, trans, ...
    blade_c_ratio_1, blade_c_ratio_2, blade_twist_1, blade_twist_2, blade_lcs_1, ...
    blade_lcs_2, x0, tip_speed_M, height, bem, k)
3 %airfoil assume NACA23-012 in root, NACA6xxxx? in tip
4
5 %% height
6 [sigma_hover, a_hover]=ISAgdp(height);
7
8 %% compute r/R, c/R, M, a, Δtheta
9 r_ratio=linspace(0,1,bem+1);
10
11 % cratio (bilinear taper)
12 % inputs: transition (centre) point @trans, taper ratio wrt c0 at root ...
    @blade_c_ratio_1 blade_c_ratio_2)
13 % trans must be at intervals of bem!
14 ccnaught=[(1-(linspace(0,1,round(trans*bem+1))).*(1-blade_c_ratio_1)), ...
    (blade_c_ratio_1-(linspace(0,1,round((1-trans)*bem+1))).*(blade_c_ratio_1-blade_c_ratio_2))];
15 ccnaught(trans*bem+1)=[];
16 cbarR=sigma*pi/blade_num;
17 cbarcnaught=(blade_c_ratio_1+1)*trans/2+(blade_c_ratio_1+blade_c_ratio_2)*(1-trans)/2;
18 c_ratio=ccnaught/cbarcnaught*cbarR;
19
20 M=r_ratio*tip_speed_M;
21
22 % blade_lcs
23 blade_lcs=[(linspace(blade_lcs_1,blade_lcs_1,round(trans*bem+1))), ...
    (linspace(blade_lcs_2,blade_lcs_2,round((1-trans)*bem+1)))];
24 blade_lcs(trans*bem+1)=[];
25
26 % twist
27 twist=[-(linspace(0,blade_twist_1,round(trans*bem+1))), ...
    -(linspace(blade_twist_1,blade_twist_2,round((1-trans)*bem+1)))];
28 twist(trans*bem+1)=[];
29
30 %% collective pitch thetanaught
31
32 pitch=-k+twist;
33
34 %% local inflow angle
35 localtan=blade_lcs.*blade_num.*c_ratio./(16.*pi.*r_ratio).*(-1+sqrt(1+(32.*pi.*pitch./180.*pi.*r_
36
37 %% local aoa
38 tanminus1=atan(localtan)./pi.*180;
39 aoa=pitch-tanminus1;
40
41 %% check graphs for cl/cd
42
43 % tiprelief = imread('tiprelief.jpg');
44 % imshow(tiprelief);
45 for i=1:round(trans*bem)
46     %fprintf('If wind tunnel testing, account for tip relief by reducing the ...
        local M number by increment read from figure. \n')
47     [cl(i),cd(i)]=NACA23012(aoa(i), M(i));
48 end
49 for i=round(trans*bem+1):(bem+1)

```

```

50     [cl(i),cd(i)]=NACA63206(aoa(i), M(i));
51 end
52
53 %% running thrust loading
54 dctdrratio=blade_num.*(r_ratio).^2.*c_ratio.*cl/(2*pi);
55
56 %% numel integration
57 ct=trapz(linspace(x0,1,100),interp1(r_ratio,dctdrratio,linspace(x0,1,100)));
58
59 %% tip loss factor
60 B=1-sqrt(2*ct)/blade_num;
61 ct=ct-trapz(linspace(B,1,100),interp1(r_ratio,dctdrratio,linspace(B,1,100)));
62
63 %% dcq0drratio
64 dcq0drratio=blade_num.*(r_ratio).^3.*c_ratio.*cd/(2*pi);
65
66 %% numel integration (for cq0 is from 0! rotor hub contributes drag)
67 cq0=trapz(linspace(0+2/(bem+1),1,100),interp1(r_ratio(2:bem+1),dcq0drratio(2:bem+1),linspace(0+2/
68
69 %% dcqidrratio
70 dcqidrratio=blade_num.*(r_ratio).^3.*c_ratio.*cl.*localtan/(2*pi);
71
72 %% numel integration
73 cqi=trapz(linspace(x0,B,100),interp1(r_ratio,dcqidrratio,linspace(x0,B,100)));
74
75 %% Δcqi due to rotation of wake
76 siptip=SIPTIP(ct);
77 Δcqi=siptip*cqi;
78
79 %% disc loading
80 DL=ct*(1.225*sigma_hover*(tip_speed_M*a_hover)^2)/(515*0.3048*0.3048); %515: ...
    slug factor, 0.3048 ft factor
81
82 %% Ct/sigma (sigma only consider 0! no x0)
83 ctsigma=ct/(sigma);
84
85 %% find mpcp
86 mpcp=MPCP(DL*ctsigma);
87
88 %% total cq
89 cq=(cq0+cqi+Δcqi)*(mpcp);
90
91 %% thrust
92 A=pi*blade_r^2;
93 T=1.225*sigma_hover*A*(tip_speed_M*a_hover)^2*ct/(515*0.3048*0.3048);
94
95 %% power
96 hp=1.225*sigma_hover*A*(tip_speed_M*a_hover)^3*cq/550/(515*0.3048*0.3048*0.3048);
97
98 %% check max aoa
99 maxaoa=max(aoa);
100
101 %% Figure of Merit
102 FM=ct^(3/2)/((sqrt(2)*cq));
103
104 end

```

D MATLAB Code for Cruise

This is the implementation of the cruise BEMT.

```

1 function [T, hp, aoa270deg, alpha, Ctsigma, Aone, Bone, iter, Cmsigma, Crsigma, ...
    Cqsigma, Chsigma, lambdaprim, M, L, alphaspp]=forward(blade_num, blade_r, ...
    sigma, trans, blade_c_ratio_1, blade_c_ratio_2, blade_twist_1, ...
    blade_twist_2, blade_lcs_1, blade_lcs_2, x0, height, Mabt, V, bem, N, ...
    fareainput, thetanaught, Aone, Bone)
2
3 %% conversions
4 omega=[0:360/N:359];
5 [sigma_forward, a_forward]=ISAgdp(height);
6 omegaR=Mabt*(a_forward/0.3048)-V;
7 %forward speed in ft/s
8 mu=V/omegaR; %TSR
9
10 blade_lcs=blade_lcs_1*trans+blade_lcs_2*(1-trans);
11
12 % twist
13 blade_twist=[-(linspace(0,blade_twist_1,round(trans*bem+1))), ...
    -(linspace(blade_twist_1,blade_twist_2,round((1-trans)*bem+1)))];
14 blade_twist(trans*bem+1)=[];
15
16 %% constants
17 g=32.174; %imperial units
18 B=0.97;
19 cf=0.006; %constant p212
20 Ab=sigma*pi*blade_r^2;
21
22 %% initialising
23 r_ratio=linspace(0,1,bem+1);
24
25 alpha=zeros(bem+1,N)';
26 UTbar=zeros(bem+1,N)';
27 UPbar=zeros(bem+1,N)';
28
29 %% guesses
30
31 Ctsigma=0.0765; %first guesses
32 Cmsigma=0.1;
33 Crsigma=0.1;
34 Cqsigma=0.1;
35 Chsigma=0.1;
36
37 %% loop
38 for iter=1:500
39
40 %% alpha
41 gamma=4; %p170/ p30, 10-lightly built 2-tip driven (for prouty, constant chord ...
    and NACA0012 - 8.1)
42 anaught=(2/3.*gamma.*Ctsigma(iter)./(blade_lcs-1.5*g*blade_r/(omegaR^2))); ...
    %contradiction: see p171/p213 trust 213
43 fareaa=fareainput/((0.3048)^2); %alessandro, 0.9m^2
44 if abs(Ctsigma(iter))<0.01
45     Ctsigma(iter)=0.01
46 end
47 alphaspp=-fareaa/Ab*(mu^2)/2/Ctsigma(iter);
48 lambdaprim(iter)=mu*(alphaspp)-Ctsigma(iter)*sigma/(2*mu);
49 UPbar=lambdaprim(iter)-Ctsigma(iter)*sigma/(2*mu).*(cos(deg2rad(omega))'*r_ratio)-(mu.*anaught.*

```

```

50 UTbar=r_ratio+mu.*(sin(deg2rad(omega)))';
51 theta=thetanaught+(ones(N,1)*(blade_twist))-Aone.*(cos(deg2rad(omega))'*ones(1,bem+1))-Bone.*(sin
    %thetanaught collective pitch %thetal tip pitch %anaught=coning %A1=lateral ...
    cyclic pitch %B1 longitudinal cyclic pitch %lambda' new inflow angle
52 Δalpha=atan2(UPbar,UTbar)/pi*180;
53 alpha=theta+Δalpha;
54
55 %% cn cl cd
56 UBbar=sqrt(UTbar.^2+UPbar.^2);
57 M=UBbar.*omegaR./(a_forward/0.3048);
58 for i=1:N
59     for j=round((trans+0.02)*bem+1):(bem+1)
60         [M(i,j)]=[M(i,j)]*cos(deg2rad(25));
61     end
62     for j=1:round(trans*bem)
63         [cl(i,j), cd(i,j)]=NACA23012(alpha(i,j),M(i,j));
64     end
65     for j=round(trans*bem+1):(bem+1)
66         [cl(i,j), cd(i,j)]=NACA63206(alpha(i,j),M(i,j));
67     end
68 end
69
70 Cn=cl.*UTbar./UBbar+cd.*UPbar./UBbar;
71
72 %% detsigmadr
73 detsigmadr=UBbar.^2./2.*Cn;
74 %% numel integration (limits include x0 root effects)
75 Δ...
    ctsigma=trapz(linspace(x0,B,100),interp1(r_ratio,detsigmadr',linspace(x0,B,100))); ...
    %(trapz(detsigmadr')/100)';
76 %% Ct/sigma first pass
77 Ctsigma(iter+1)=(1/(length(omega))*sum(Δctsigma)); %Ctsigma(iter)/2+0.5*
78 % B_t=1-sqrt(abs(2*Ctsigma*solidity))/blade_num;
79 % %% numel int (include B tip effects)
80 % Δ...
    ctsigma=Δctsigma-trapz(linspace(B_t,1,100),interp1(r_ratio_ini,detsigmadr',linspace(B_t,1,100)));
    %remember now used average B rather than for each angle omega!
81 % Ctsigma=1/(length(omega))*sum(Δctsigma);
82
83 %% dcmsigmadr
84 dcmsigmadr=(-UBbar.^2./2.*Cn).*(cos(deg2rad(omega))'*r_ratio);
85 %% numel integration (limits include x0 root effects)
86 Δ...
    cmsigma=trapz(linspace(x0,B,100),interp1(r_ratio,dcmsigmadr',linspace(x0,B,100))); ...
    %(trapz(dcmsigmadr')/100)';
87 %% Cm/sigma first pass
88 Cmsigma(iter+1)=(1/(length(omega))*sum(Δcmsigma)); %Cmsigma(iter)/2+0.5*
89
90
91 %% dcrsigmadr
92 dcrsigmadr=(-UBbar.^2./2.*Cn).*(sin(deg2rad(omega))'*r_ratio);
93 %% numel integration (limits include x0 root effects)
94 Δ...
    crsigma=trapz(linspace(x0,B,100),interp1(r_ratio,dcrsigmadr',linspace(x0,B,100))); ...
    %(trapz(dcrsigmadr')/100)';
95 %% Cr/sigma first pass
96 Crsigma(iter+1)=(1/(length(omega))*sum(Δcrsigma)); %Crsigma(iter)/2+0.5*
97
98 %% SF
99 URbar=mu.*(cos(deg2rad(omega))'*ones(1,bem+1));
100 UR=URbar.*omegaR;

```

```

101
102 UT=UTbar.*omegaR;
103
104 UB=UBbar.*omegaR;
105
106 ccsf=UT.*sqrt(UT.^2+UR.^2)./(UB.^2).*cf;
107 cdp=cd-0.006;
108 cco=cdp.*UTbar./UBbar+cf.*UTbar.*sqrt(UTbar.^2+URbar.^2)./(UBbar.^2);
109
110 ccimd=-cl.*UPbar./UBbar;
111
112 cc=cco+ccimd;
113
114 %% dcqsigmadrr
115 dcqsigmadrr=UBbar.^2./2.*(ones(N,1)*r_ratio).*cc;
116
117 %% dchsigmadrr
118 dchsigmadrr=UBbar.^2./2.*((cc.*(sin(deg2rad(omega))'*ones(1,bem+1)))+(cf.*mu./UTbar).*(cos(deg2rad(omega))'*ones(1,bem+1))));
119
120 %% dont apply root cutout or tip loss to pressure and skin friction! (to be ...
    done, separate in lift and non lift related)
121
122 %% numel integration
123 Δcqsigma=trapz(linspace(x0,B,100),interp1(r_ratio,dcqsigmadrr',linspace(x0,B,100)));
124 Δchsigma=trapz(linspace(x0,B,100),interp1(r_ratio,dchsigmadrr',linspace(x0,B,100)));
125 %% Cq/sigma
126 Cqsigma(iter+1)=(1/(length(omega))*sum(Δcqsigma)); %Cqsigma(iter)/2+0.5*
127 Chsigma(iter+1)=(1/(length(omega))*sum(Δchsigma)); %Chsigma(iter)/2+0.5*
128
129 %% Aone/Bone
130 Aone=Aone-(60*(2+mu.^2)*Cmsigma(iter+1));
131 Bone=Bone-(190*(1-mu)*Crsigma(iter+1));
132
133 %% exit criteria
134 if abs(Crsigma(iter+1)-Crsigma(iter))<0.005 && ...
    abs(Cmsigma(iter+1)-Cmsigma(iter))<0.002
135 break
136 end
137
138 iter;
139 if iter>15
140     error('does not converge')
141 end
142 end
143
144
145
146
147 T=1.225*sigma_forward*Ab*(omegaR)^2*Ctsigma(end)/(515); %T=GW!
148 Q=1.225*sigma_forward*Ab*(omegaR)^2*Cqsigma(end)/(515)*blade_r; %torque (Cp=Cq)
149 hp=1.225*sigma_forward*Ab*(omegaR)^3*Cqsigma(end)/(515)/550; %Power in hp
150 aoa270deg=alpha(floor(N/4*3+1),end);
151
152 L=T*cos(deg2rad(abs(alphatpp)));
153 end

```

E MATLAB Code for Isight

This is the MATLAB code used in Isight to pass on input parameters and calculate output parameters.

```

1  %% established inputs
2
3  blade_num=2 %baseline
4  blade_r=13.2 %ft, iteration with hover
5  AR=14 %aero choice, justify with struc
6  sigma=2*blade_num/(AR*pi)
7  x0=0.15 %justify with system
8  height=20000
9  [sigma_height,a_height]=ISAgdp(height)
10 bem=50
11 Mabt=0.85 %decided aero
12 V=205.05 %ft/s, 225km/h dashspeed
13 tip_speed_M=(0.85*(a_height/0.3048)-V)/(a_height/0.3048)
14 N=8
15 Aone=0
16 Bone=0
17
18 blade_lcs_1=[6.8153]
19 blade_lcs_2=[7.0268]
20 farea=[0.418] %positive constant from alessandro, m^2
21
22 %% match
23 desiredDL=9.4 %lbf/ft^2
24 theta=14 %inclination of rotor
25 W=2900 %kg
26 desiredT=W/2*9.81/4.448/cos(deg2rad(theta))
27 overlap=0.3454
28 DLfactor=2/(1+overlap)
29
30 desiredfinalL=desiredT
31 desiredfinalT=desiredT
32 desiredfinalDL=desiredDL*DLfactor
33
34 [Th, hp, maxaoa, DL, FM]=hover(blade_num, blade_r, sigma, trans, ...
    blade_c_ratio_1, blade_c_ratio_2, blade_twist_1, blade_twist_2, blade_lcs_1, ...
    blade_lcs_2, x0, tip_speed_M, height, bem, k);
35 [Tf, hp, aoa270deg, alpha, Ctsigma, ̳, ̳, iter, Cmsigma, Crsigma, Cqsigma, ...
    Chsigma, lambdaprime, M, L]=forward(blade_num, blade_r, sigma, trans, ...
    blade_c_ratio_1, blade_c_ratio_2, blade_twist_1, blade_twist_2, blade_lcs_1, ...
    blade_lcs_2, x0, height, Mabt, V, bem, N, farea, thetanaught, Aone, Bone);
36 LDR= (W/2*9.81/4.448*V)/(hp*550);

```